

Inventory & Supply Chain Analytics Project – Insights Report

1. Introduction

Effective inventory management plays a critical role in maintaining product availability, minimizing operational costs, and supporting efficient supply chain operations. This project analyzes inventory data from a retail environment to identify inventory patterns, supplier contributions, product movement trends, and demand forecasting opportunities.

The analysis was performed using Microsoft Excel and includes inventory classification, category-level performance analysis, supplier evaluation, inventory turnover assessment, demand trend analysis, and forecasting techniques to support data-driven inventory decisions.

2. Project Objectives

The primary objectives of this project were:

- To classify inventory items based on their relative importance using ABC Analysis
- To analyze inventory distribution across product categories
- To evaluate supplier contribution to inventory availability
- To identify fast-moving products using inventory turnover analysis
- To examine monthly demand patterns for key products
- To generate demand forecasts using a 3-Month Moving Average method
- To develop an interactive inventory dashboard for management reporting

3. Dataset Overview

The dataset contains inventory records across multiple product categories, suppliers, and stock levels. Key data fields include:

- Item ID
- Item Name
- Category
- Supplier

- Stock Available
- Units Sold
- Inventory Turnover
- Stock Status
- Monthly Demand

The dataset consists of 40 inventory items spanning Grocery, Dairy, Household, Personal Care, Fruits, Vegetables, Beverages, Snacks, Bakery, Frozen Foods, and Packaged Food categories.

4. ABC Analysis

Key Insights

- A-category items account for the highest inventory movement and represent the most critical products for inventory management
- B-category items contribute moderate inventory movement and require periodic monitoring
- C-category items contribute the lowest inventory movement and require relatively less operational attention
- High-demand categories such as Vegetables, Dairy, and Beverages are primarily concentrated within A-category items

Findings

- A-category products have the greatest impact on inventory availability and service levels
- Inventory management efforts should be prioritized toward A-category items
- B-category products require balanced replenishment policies
- C-category items can be managed with less frequent review cycles

Key Takeaway

Effective inventory control should focus primarily on A-category items, as they drive the majority of inventory movement and have the greatest impact on stock availability.

Summary

ABC analysis demonstrates that inventory priorities are not evenly distributed across products. Prioritizing high-value and high-movement items can improve inventory efficiency and reduce stockout risks.

5. Category-Wise Inventory Analysis

Key Insights

- Grocery holds the highest inventory volume with 370 units
- Household and Personal Care categories each contribute 300 units to total inventory
- Fruits, Vegetables, and Packaged Food maintain comparatively lower stock levels
- Inventory distribution varies significantly across product categories

Findings

- Inventory investment is concentrated in a small number of major categories
- High-stock categories require regular monitoring to prevent excess inventory accumulation
- Lower-stock categories may require closer review to avoid stock shortages
- Category performance differs according to demand patterns

Key Takeaway

Inventory is heavily concentrated in a small number of categories, making category-level monitoring essential for balanced stock management.

Summary

Category analysis provides visibility into inventory allocation across product groups and supports informed replenishment and inventory planning decisions.

6. Supplier Performance Analysis

Key Insights

- AgroFresh Supplies Pvt Ltd contributes the largest inventory volume
- HygieneCorp Suppliers is the second-largest supplier by inventory contribution
- Inventory procurement is concentrated among a few major suppliers

- Supplier contributions vary significantly across the supplier network

Findings

- Dependence on a limited number of suppliers may increase supply chain risk
- Major suppliers play a critical role in maintaining inventory availability
- Supplier performance can be evaluated based on inventory contribution levels
- Supplier diversification may improve supply chain resilience

Key Takeaway

A small number of suppliers account for a substantial share of inventory, highlighting the importance of supplier relationship management and procurement planning.

Summary

Supplier analysis helps identify key procurement partners and supports data-driven sourcing and supplier management decisions.

7. Inventory Turnover Performance Analysis

Key Insights

- Bread exhibits the highest inventory turnover ratio among all analyzed products
- Milk and Potatoes also demonstrate strong inventory movement
- Fast-moving products require more frequent replenishment
- Inventory turnover analysis helps identify priority inventory items

Findings

- High-turnover items experience faster inventory consumption
- Replenishment priorities should align with turnover performance
- Lower-turnover products may require periodic review to avoid excess inventory holding
- Inventory movement varies considerably across products

Key Takeaway

Fast-moving products should receive priority replenishment attention to maintain product availability and support efficient inventory operations.

Summary

Inventory turnover analysis provides insight into product movement patterns and supports inventory optimization through targeted replenishment planning.

8. Monthly Demand Analysis (Potatoes)

Key Insights

- Potato demand decreased from 25 units in January to 10 units in June
- The 3-Month Moving Average forecast estimates July demand at approximately 10 units
- Actual demand closely follows the forecast trend during later months
- Demand patterns indicate a transition from high consumption levels to a more stable demand environment

Findings

- Demand declined by approximately 60% during the six-month period
- The decline was gradual and consistent rather than a short-term fluctuation
- Forecasted demand suggests no immediate increase in inventory requirement
- Historical demand data supports a lower replenishment strategy

Key Takeaway

Potato demand has stabilized at approximately 10 units per month after a period of decline, supporting more conservative inventory replenishment decisions.

Summary

The analysis demonstrates a sustained reduction in potato demand, with forecasting results indicating stable future demand levels.

9. Forecast Analysis

Key Insights

- Potatoes recorded the highest cumulative demand, followed closely by Onions
- Fruit Juice and Handwash demonstrated relatively stable demand patterns
- Most fast-moving products experienced lower demand in May and June compared to the first quarter

- Demand patterns highlight the importance of monitoring high-priority inventory items

Findings

- The selected fast-moving products generated a significant share of total demand
- Potatoes and Onions contributed the largest share of demand among the analyzed items
- Demand variability differs across products, requiring item-specific replenishment strategies
- All selected products remained high-priority inventory items throughout the analysis period

Key Takeaway

Potatoes and Onions remain the most critical fast-moving products, while stable-demand products provide predictable replenishment opportunities.

Summary

Demand forecasting supports proactive inventory planning, helping maintain product availability while minimizing excess inventory.

10. Executive Dashboard Summary

The Executive Dashboard consolidates inventory performance metrics into a single reporting view. Interactive slicers allow users to analyze inventory by category, supplier, and stock status.

Key dashboard metrics include:

- Total Inventory Items
- Total Stock Available
- Reorder Status Monitoring
- Category-Level Inventory Distribution
- Supplier Contribution Analysis
- Top Fast-Moving Product Items

The dashboard enables quick identification of inventory priorities and supports operational decision-making.

11. Overall Recommendations

- To prioritize monitoring and replenishment of A-category inventory items
- To review high-turnover products more frequently to prevent stockouts
- To monitor supplier concentration and evaluate alternative sourcing opportunities
- To use demand forecasting to align inventory replenishment with consumption trends
- To regularly review inventory distribution across categories to maintain optimal stock levels

12. Conclusion

This project demonstrates how inventory analytics can be applied to improve inventory visibility, replenishment planning, supplier management, and demand forecasting. By leveraging Excel-based analytical techniques such as ABC Analysis, inventory turnover evaluation, and demand forecasting, organizations can make more informed inventory decisions, reduce operational inefficiencies, and improve overall supply chain performance.